

Solution Set 9

① Let us solve the differential equation

①

$$[\nabla^2 + k^2] G(\vec{r}) = \delta^3(\vec{r})$$

We see that except at the origin the above equation is the same as the equation one would solve for the wave function of free particles!

$$[\nabla^2 + k^2] G(\vec{r}) = 0 \quad \text{for } |\vec{r}| \neq 0.$$

We know that the general solution to this equation (in spherical polar coordinates) is given by

$$G(\vec{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l [A_l j_l(kr) + B_l n_l(kr)] Y_l^m(\theta, \phi)$$

where $j_l(kr)$ and $n_l(kr)$ are spherical Bessel and spherical Neumann functions.

Now for $r \rightarrow \infty$ we want the "radial" part of the Green's function to be of the form $\frac{e^{ikr}}{r}$.

We know for $r \rightarrow \infty$

$$j_l(kr) \rightarrow \frac{1}{kr} \sin(kr - \frac{l\pi}{2})$$

$$n_l(kr) \rightarrow -\frac{1}{kr} \cos(kr - \frac{l\pi}{2})$$

Thus we need $A_l = -B_l$ to cancel the $\frac{e^{-ikr}}{r}$ part. Thus

$$G(\vec{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_l [j_l(kr) - n_l(kr)] Y_l^m(\theta, \phi).$$

Now $\delta^3(\vec{r})$ has an integrable singularity as $r \rightarrow 0$! which means $\delta^3(\vec{r}) \rightarrow \frac{1}{r^2}$ as $r \rightarrow 0$! This means the coefficient of $\eta_l(kr)$ must be zero for all $l > 0$! (2)

Remember

$$\eta_l(kr) \sim \frac{1}{(kr)^{l+1}}$$

$$\text{and } \nabla^2 \eta_l(kr) \sim \frac{1}{(kr)^{l+2}}$$

Hence

$$G(\vec{r}) = A_0 \frac{e^{ikr}}{kr}$$

To determine A_0 we integrate the original eq. over V (a small sphere of radius ε) i.e. ...

$$\int_V d^3r [\nabla^2 + k^2] G(\vec{r}) = \int_V d^3r \delta^3(\vec{r})$$

Substituting we get

$$\begin{aligned} \int_V d^3r \vec{\nabla} \cdot \vec{\nabla} G(\vec{r}) + k^2 \int_V d^3r G(\vec{r}) &= 1 \\ &\quad \searrow = 0 \text{ (taking } \varepsilon \rightarrow 0) \\ = \int_S d\Omega \hat{e}_r \cdot \vec{\nabla} G(\vec{r}) &= 1 \end{aligned}$$

$\uparrow$
 Surface of the sphere

$$\Rightarrow A_0 = -\frac{k}{4\pi} \Rightarrow$$

$$G(\vec{r}) = -\frac{1}{4\pi} \frac{e^{ikr}}{r}$$

② The Born approximation is valid as long as

③

$$\frac{2\mu}{\hbar^2 k} \left| \int_0^{\infty} e^{ikr'} \sin kr' V(r') dr' \right| \ll 1.$$

for a spherically symmetric potential. For the Yukawa potential this condition becomes

$$\frac{2\mu |V_0|}{\hbar^2 k \alpha} \left| \int_0^{\infty} e^{ikr'} \sin kr' \frac{e^{-\alpha r'}}{r'} dr' \right| \ll 1$$

When $\frac{k}{\alpha} \ll 1$ we can write the above condition as

$$\frac{2\mu V_0}{\hbar^2 k \alpha} \left| \int_0^{\infty} e^{i \frac{k}{\alpha} t} \sin \frac{k}{\alpha} t e^{-t} \frac{dt}{t} \right| \ll 1$$

where we substituted $dr' = t$

which then leads to

$$\boxed{\frac{2\mu |V_0|}{\hbar^2 \alpha^2} \ll 1} \quad (\text{Low energies}).$$

When $\frac{k}{\alpha} \gg 1$ we can write the condition as

$$\frac{2\mu |V_0|}{\hbar^2 k \alpha} \left| \int_0^{\infty} e^{it} \frac{\sin t}{t} e^{-\frac{\alpha}{k} t} dt \right|$$

Now

(4)

$$\int_0^{\infty} e^{it} \frac{\sin t}{t} e^{-\frac{\alpha}{k}t} dt = \int_0^{\infty} \left\{ \frac{1}{2} \frac{\sin 2t}{t} e^{-\frac{\alpha}{k}t} + i \frac{\sin^2 t}{t} e^{-\frac{\alpha}{k}t} \right\} dt$$

From the "table of integrals" we see that-
(Gradshteyn & Ryzhik)

$$\int_0^{\infty} \frac{\sin 2t}{t} e^{-\frac{\alpha}{k}t} dt = \tan^{-1}\left(\frac{2k}{\alpha}\right)$$

$$\int_0^{\infty} \frac{\sin^2 t}{t} e^{-\frac{\alpha}{k}t} dt = \frac{1}{4} \ln\left(\frac{(\frac{\alpha}{k})^2 + 4}{(\frac{\alpha}{k})^2}\right)$$

In the $\frac{\alpha}{k} \rightarrow 0$ limit we see that-

$$\int_0^{\infty} e^{it} \frac{\sin t}{t} e^{-\frac{\alpha}{k}t} dt \approx -\frac{i}{2} \ln\left(\frac{\alpha}{k}\right) \approx \frac{i}{2} \ln\left(\frac{k}{\alpha}\right)$$

Also
* See the end of Problem

$\therefore$ The condition is given by

$$\frac{\mu |V_0|}{\hbar^2 k \alpha} \ln\left(\frac{k}{\alpha}\right) \ll 1$$

You can roughly estimate the above integral as follows:

$$\left\{ \int_0^{\infty} \frac{e^{it} \sin t}{t} e^{-\frac{\alpha}{k}t} dt \approx \int_0^{k/\alpha} \frac{e^{it} \sin t}{t} dt \xrightarrow[\text{for } k/\alpha \rightarrow \infty]{\text{diverges as } \ln\left(\frac{k}{\alpha}\right)} \right\}$$

(5)

When there are bound states

$$\frac{2\mu|V_0|}{\hbar^2 \alpha^2} \geq 2.7.$$

Thus the Born approximation is violated at

low energies but not ^{necessarily} at high energies! This is reasonable since the bound states affect only low energies.

Thus the sign of V_0 should not matter at high energies!

* We can also estimate the integral $\int_0^\infty e^{it} \frac{\sin t}{t} e^{-\frac{\alpha}{k}t} dt$ more easily! Since the exponential is simply cutting off the t integral we can approx it by

$$\approx \int_0^{k/\alpha} e^{it} \frac{\sin t}{t} dt \approx \int_0^{k/\alpha} \frac{e^{2it}}{2it} - \int_0^{k/\alpha} \frac{1}{2it} dt$$

We only want the singular part as $k/\alpha \rightarrow \infty$. This is obtained from the 2nd integral (first one is convergent at any k/α due to oscillating part). The second integral is $\frac{1}{2} \ln k/\alpha$ as expected!

(3) The Lippmann-Schwinger formalism in 1-d can be formulated as follows:

We want to solve

$$\hat{H} |\psi_{E_k}\rangle = E_k |\psi_{E_k}\rangle$$

where

$$\hat{H} = \hat{H}_0 + \hat{V}$$

we know $\hat{H}_0 |\phi_k\rangle = E_k |\phi_k\rangle$ (Free problem).

We want to solve

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$$|\psi_{E_k}\rangle = |\phi_{E_k}\rangle + |\psi_{sc}\rangle$$

$$\begin{aligned} \cancel{H_0 |\phi_{E_k}\rangle} + V |\phi_{E_k}\rangle + H_0 |\psi_{sc}\rangle + V |\psi_{sc}\rangle \\ = \cancel{E_k |\phi_{E_k}\rangle} + E_k |\psi_{sc}\rangle \end{aligned}$$

$$(E_k - H_0) |\psi_{sc}\rangle = V |\phi_{E_k}\rangle$$

$$|\psi_{sc}\rangle = \frac{1}{E_k - H_0} V |\phi_{E_k}\rangle$$

$$|\psi_{E_k}\rangle = |\phi_{E_k}\rangle + \frac{1}{E_k - H_0} V |\phi_{E_k}\rangle$$

Thus we need to find the prescription for handling the singularity in $\frac{1}{E_k - H_0}$.

In 1-d we have

$$\psi_{E_k}(x) = \phi_{E_k}(x) + \langle x | \frac{1}{E_k - H_0} \hat{V} | \psi_{E_k} \rangle$$

Let us assume that the prescription $E_k \rightarrow E_k + i\epsilon$ works and find out if the Green's function

$$G^{(+)}(x, x') = \frac{\hbar^2}{2m} \langle x | \frac{1}{E_k - H_0 + i\epsilon} | x' \rangle$$

has the right boundary condition.

(7)

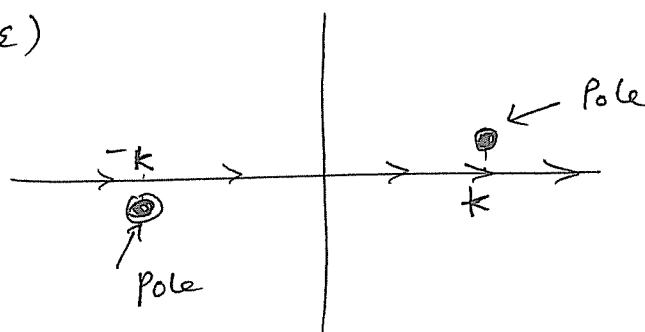
Now $G^{(+)}(x, x') = \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \frac{dp}{2\pi\hbar} e^{\frac{i}{\hbar} p(x-x')} \frac{1}{(E_k - \frac{p^2}{2m} + i\varepsilon)}$

Let us substitute $\frac{p}{\hbar} = q$ and $E_k = \frac{\hbar^2 k^2}{2m}$. Then we get

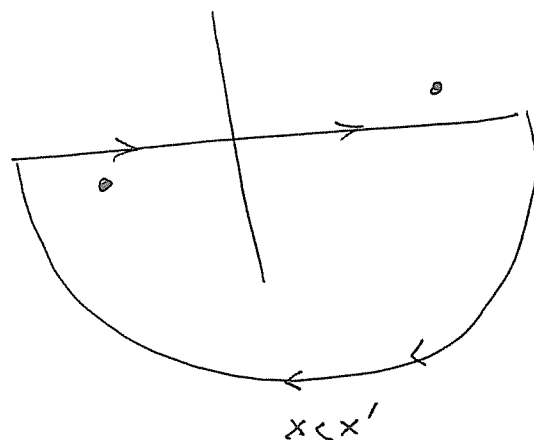
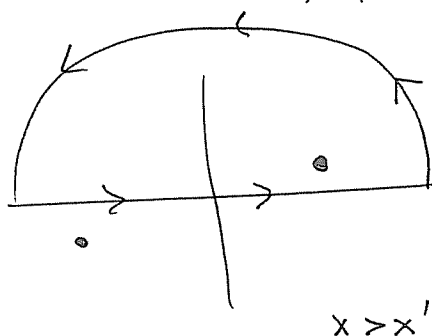
$$G^{(+)}(x, x') = - \int_{-\infty}^{\infty} \frac{dq}{2\pi} \frac{e^{iq(x-x')}}{(q^2 - k^2 - i\varepsilon)}$$

The poles are at

$$q = \pm (k + i\varepsilon')$$



Now if $x > x'$ we can close the contour in the upper half plane and when $x < x'$ we can close the contour in the lower half plane.



Thus

$$G^{(+)}(x, x') = - \oint \frac{dq}{2\pi} \frac{e^{iq(x-x')}}{(q - k - i\varepsilon')(q + k + i\varepsilon')}$$

$$= \begin{cases} -\frac{i}{2k} e^{ik(x-x')} & x > x' \\ -\frac{i}{2k} e^{-ik(x-x')} & x < x' \end{cases}$$

Thus

$$G(x, x') = \frac{-i}{2k} e^{ik|x-x'|}$$

Thus the integral equation for $\psi_k^+(x)$ is

$$\psi_k^+(x) = \psi_{inc}^+(x) + \int dx' G(x, x') V(x') \psi_k^+(x')$$

(b) Now we are given $V(x)$, so

$$\psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + \frac{2m}{\hbar^2} \int dx' \left(-\frac{i}{2k} e^{ik|x-x'|} \right) \left(-\frac{\gamma \hbar^2}{2m} \right) \delta(x') \psi_k^+(x')$$

$$\therefore \psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + \frac{i\gamma}{2k} e^{ik|x|} \psi_k^+(0)$$

Now substituting $x=0$ we see

$$\psi_k^+(0) \left[1 - \frac{i\gamma}{2k} \right] = \frac{1}{\sqrt{2\pi}}$$

$$\therefore \psi_k^+(0) = \frac{1}{\sqrt{2\pi}} \left(1 - \frac{i\gamma}{2k} \right)^{-1}$$

Hence we see

$$\psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + \frac{i\gamma}{2k} \frac{1}{\sqrt{2\pi}} \left(1 - \frac{i\gamma}{2k} \right)^{-1} e^{ik|x|}$$

for $x \rightarrow +\infty$ we see

$$\psi_k^+(x) = \frac{1}{\sqrt{2\pi}} \left(1 + \frac{\frac{i\gamma}{2k}}{1 - \frac{i\gamma}{2k}} \right) e^{ikx}$$

for $x \rightarrow -\infty$ we see

(9)

$$\psi_k^+(x) = \frac{1}{\sqrt{2\pi}} e^{ikx} + \frac{1}{\sqrt{2\pi}} \frac{\frac{i\gamma}{2k}}{\left(1 - \frac{i\gamma}{2k}\right)} e^{-ikx}$$

$$\therefore R = \left| \frac{\frac{i\gamma}{2k}}{\left(1 - \frac{i\gamma}{2k}\right)} \right|^2 = \left(\frac{\gamma^2}{4k^2 + \gamma^2} \right)$$

$$\text{and } T = \left| \frac{1}{1 - \frac{i\gamma}{2k}} \right|^2 = \left(\frac{4k^2}{4k^2 + \gamma^2} \right)$$

Exercise 19.5.6 (Optical Theorem)

(1) The wavefunction is given by

$$\psi(\vec{r}) = \psi_{inc} + \psi_{sc}$$

where

$$\left. \begin{aligned} \psi_{inc} &= e^{ikr \cos \theta} \\ \psi_{sc} &= \frac{e^{ikr}}{r} f(\theta) \end{aligned} \right\} \text{ for } r \rightarrow \infty.$$

The current density due to the interference term is given by

$$(\vec{j})^{int} = \frac{\hbar}{2\mu i} \left[\psi_{inc}^* \vec{\nabla} \psi_{sc} + \psi_{sc}^* \vec{\nabla} \psi_{inc} - \vec{\nabla} \psi_{inc}^* \psi_{sc} - \vec{\nabla} \psi_{sc}^* \psi_{inc} \right].$$

$$= \frac{\hbar}{\mu} \text{Im} \left[\psi_{inc}^* \vec{\nabla} \psi_{sc} + \psi_{sc}^* \vec{\nabla} \psi_{inc} \right].$$

If we look at the radial part at $r \rightarrow \infty$ we just

consider

$$\vec{\nabla} \psi_{sc} \rightarrow \frac{\partial}{\partial r} \psi_{sc} \quad \text{and} \quad \vec{\nabla} \psi_{inc} \rightarrow \frac{\partial}{\partial r} \psi_{inc}$$

Substituting we get -

$$j_r^{int} \underset{r \rightarrow \infty}{\sim} \frac{\hbar k}{\mu r} \text{Im} \left[i e^{ikr(1-\cos \theta)} f(\theta) + e^{ikr(\cos \theta - 1)} f^*(\theta) \cos \theta \right]$$

(2) As long as $\theta \neq 0$ we see that $(1 - \cos\theta)$ is not zero.

This means $\int_r^{\text{int}}$ oscillates wildly over any small change in the angle and hence integrates to zero!
 (when $r \rightarrow \infty$)

(3) When $\theta \simeq 0$ $(1 - \cos\theta) \simeq \frac{\theta^2}{2}$. This means

$$\int \underbrace{j_r^{\text{int}}}_{\substack{\text{forward} \\ \text{cone} \\ \downarrow \\ \theta = 0 \text{ to } \theta \text{ (small)} \\ \phi = 0 \text{ to } 2\pi}} r^2 d\Omega \simeq \left(\frac{\hbar k}{u}\right) \cdot r \cdot \text{Im} \left[i \int_0^{2\pi} \int_0^\theta e^{-ikr\theta^2/2} \theta d\theta d\phi f^*(0) + i \int_0^{2\pi} \int_0^\theta e^{ikr\theta^2/2} \theta d\theta d\phi f(0) \right]$$

$$\simeq \left(\frac{\hbar k}{u}\right) r 2\pi \text{Im} \left[i f^*(0) \left(\frac{e^{-ikr\theta^2/2}}{ikr} \right)_0^\theta + i f(0) \left(\frac{e^{ikr\theta^2/2}}{ikr} \right)_0^\theta \right]$$

Now in the $r \rightarrow \infty$ limit we can set $\frac{\pm ikr\theta^2}{2} \rightarrow 0$
 (using $\nearrow$ argument in Shankar page 66 on evaluating rapidly oscillating functions).

$$\simeq -\left(\frac{\hbar k}{u}\right) \cdot \frac{4\pi}{k} \text{Im} f(0)$$

Exercise 19.6.1 (1) We know from (19.6.16) and (19.6.17) that-

$$\theta_L = \theta_{cm}/2 \quad \phi_L = \phi_{cm}.$$

Further

$$\left(\frac{d\sigma}{d\Omega_L} \right) \sin\theta_L d\theta_L d\phi_L = \left(\frac{d\sigma}{d\Omega_{cm}} \right) \sin\theta_{cm} d\theta_{cm} d\phi_{cm}.$$

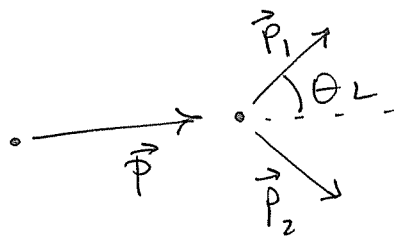
From the transformation above we see

$$\left(\frac{d\sigma}{d\Omega_L} \right) = \left(\frac{d\sigma}{d\Omega_{cm}} \right) \cdot \frac{\sin\theta_{cm}}{\sin\theta_L} \cdot \left[\frac{(d\theta_{cm} d\phi_{cm})}{(d\theta_L d\phi_L)} \right]$$

↑
Jacobian of the transformation

$$\left(\frac{d\sigma}{d\Omega_L} \right) = \left(\frac{d\sigma}{d\Omega_{cm}} \right) \bigg|_{\substack{\theta_{cm}=2\theta_L \\ \phi_{cm}=\phi_L}} \cdot 4 \cos\theta_L$$

(2) Let $\vec{P}$ be the momentum of the projectile before collision.



Let $\vec{P}_1$ be the momentum of the projectile after collision
and $\vec{P}_2$ be the momentum of the target after collision.

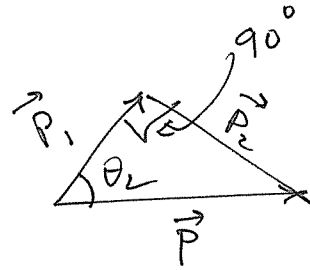
From momentum conservation

$$\vec{P} = \vec{P}_1 + \vec{P}_2.$$

From Energy (kinetic Energy) conservation (Non-relativistic)

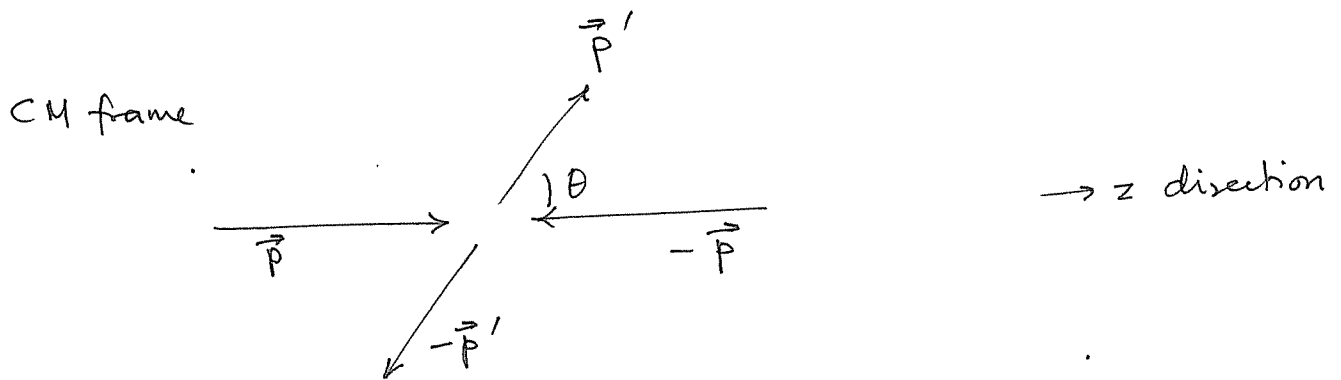
$$P^2 = P_1^2 + P_2^2.$$

Thus $\vec{P}_1$, $\vec{P}_2$ and $\vec{P}$ form a right triangle as shown below



clearly $0 \leq \theta_L \leq \pi/2$!

(3) For scattering of unequal masses.



$$\tan \theta = \frac{\sqrt{P_x'^2 + P_y'^2}}{P_z'} \quad \tan \phi = \frac{P_y'}{P_x'} \quad P = \sqrt{P_x'^2 + P_y'^2 + P_z'^2}$$

The velocity of the target = $-\frac{\vec{P}}{m_2}$ in CM frame

In the Lab frame

$$\vec{P}_{\text{proj}} = \vec{P} + \frac{m_1}{m_2} \vec{P}$$

$$\vec{P}_{\text{target}} = -\vec{P} + \frac{m_2}{m_2} \vec{P} = 0$$

$$\vec{P}'_{\text{proj}} = \vec{P}' + \frac{m_1}{m_2} \vec{P}$$

$$\vec{P}'_{\text{target}} = -\vec{P}' + \vec{P}$$

$$\left. \begin{aligned} (\vec{P}'_{\text{proj}})_x &= P_x' \\ (\vec{P}'_{\text{proj}})_y &= P_y' \\ (\vec{P}'_{\text{proj}})_z &= P_z' + \frac{m_1}{m_2} P_- \end{aligned} \right\}$$

which means

$$\tan \theta_L = \frac{\sqrt{(\vec{P}'_{\text{proj}})_x^2 + (\vec{P}'_{\text{proj}})_y^2}}{(\vec{P}'_{\text{proj}})_z}$$

$$\tan \theta_L = \frac{\sqrt{P_x'^2 + P_y'^2}}{P_z' + \frac{m_1}{m_2} P} = \frac{\sin \theta}{\cos \theta + \frac{m_1}{m_2}}$$

Exercise 19.6.3 Consider two fermions (spin- $1/2$) in the triplet state scattering off each other. Then the wave function will be antisymmetric in space. This means.

$$\frac{d\sigma}{d\Omega} = \left| f(\theta, \phi) - f(\pi - \theta, \phi + \pi) \right|^2$$

If $f(\theta, \phi) \approx f(\theta)$ then.

$$\frac{d\sigma}{d\Omega} = \left| f(\theta) - f(\pi - \theta) \right|^2$$

clearly at $\theta = \pi/2$ $\left(\frac{d\sigma}{d\Omega} \right) = 0!$